Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

---

# 8 Proofs Involving Sets

**Use the methods introduced in this chapter to prove the following statements.**

**2. Prove that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.**

*Proof.* First we show $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Suppose $a \in \{6n : n \in \mathbb{Z}\}$. This means $a = 6n$ for some integer $n$. Therefore $a = 2(3n)$ and $a = 3(2n)$. From $a = 2(3n)$, it follows that $a$ is a multiple of 2, so $a \in \{2n : n \in \mathbb{Z}\}$. From $a = 3(2n)$, it follows that $a$ is a multiple of 3, so $a \in \{3n : n \in \mathbb{Z}\}$. Thus by definition of intersection of two sets, we have $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. Therefore $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$.

Next we show $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.
Suppose $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. By definition of intersection of two sets, this means $a \in \{2n : n \in \mathbb{Z}\}$ and $a \in \{3n : n \in \mathbb{Z}\}$. Since $a \in \{2n : n \in \mathbb{Z}\}$, we know $a = 2k$ for some $k \in \mathbb{Z}$. Thus $a$ is even. Since $a \in \{3n : n \in \mathbb{Z}\}$, we know $a = 3l$ for some $l \in \mathbb{Z}$. Since 3 is odd, if $l$ were odd then $3l$ would be odd; as $a = 3l$ is even, it follows that $l$ is even. Then $l = 2j$ for some $j \in \mathbb{Z}$. So we have $a = 3l = 3(2j) = 6j$. Since $j \in \mathbb{Z}$, it follows that $a \in \{6n : n \in \mathbb{Z}\}$. Thus $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$.

Therefore, since $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ and $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$, it follows that $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$. □

**4. If $m, n \in \mathbb{Z}$, then $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$.**

*Proof.* Suppose $a \in \{x \in \mathbb{Z} : mn \mid x\}$. Then $a = mnk$ for some $k \in \mathbb{Z}$. So we have $a = m(nk)$ and $a = n(mk)$. By definition of divisibility, $m \mid a$ and $n \mid a$. Thus $a \in \{x \in \mathbb{Z} : m \mid x\}$ and $a \in \{x \in \mathbb{Z} : n \mid x\}$. By definition of intersection of two sets, we have $a \in \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. Therefore $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$. □

**6. Suppose $A, B$ and $C$ are sets. Prove that if $A \subseteq B$, then $A - C \subseteq B - C$.**

*Proof.* We prove the contrapositive: if $A - C \not\subseteq B - C$, then $A \not\subseteq B$. Assume $A - C \not\subseteq B - C$. Then there exists an element $x \in A - C$ such that $x \notin B - C$. Since $x \in A - C$, we have $x \in A$ and $x \notin C$. Since $x \notin B - C$, it follows that $x \notin B$ or $x \in C$. Because $x \notin C$, it follows $x \notin B$. Since $x \in A$ and $x \notin B$, by definition of a subset, we have $A \not\subseteq B$. □

**8. If $A, B$ and $C$ are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
A \cup (B \cap C) &= \{x : x \in A \vee x \in B \cap C\} && \text{(definition of } \cup) \\
&= \{x : x \in A \vee (x \in B \wedge x \in C)\} && \text{(definition of } \cap) \\
&= \{x : (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{(distributive law)} \\
&= \{x : x \in A \cup B \wedge x \in A \cup C\} && \text{(definition of } \cup) \\
&= \{x : x \in (A \cup B) \cap (A \cup C)\} && \text{(definition of } \cap) \\
&= (A \cup B) \cap (A \cup C) && \text{(definition of set-builder)}
\end{aligned}$$

□

**10. If $A$ and $B$ are sets in a universal set $U$, then $\overline{A \cap B} = \overline{A} \cup \overline{B}$.**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
\overline{A \cap B} &= \{x \in U : x \notin A \cap B\} && \text{(definition of complement)} \\
&= \{x \in U : \neg(x \in A \wedge x \in B)\} && \text{(definition of } \cap) \\
&= \{x \in U : x \notin A \vee x \notin B\} && \text{(DeMorgan's law)} \\
&= \{x \in U : x \in \overline{A} \vee x \in \overline{B}\} && \text{(definition of complement)} \\
&= \{x \in U : x \in \overline{A} \cup \overline{B}\} && \text{(definition of } \cup) \\
&= \overline{A} \cup \overline{B} && \text{(definition of set-builder)}
\end{aligned}$$

□

**12. If $A, B$ and $C$ are sets, then $A - (B \cap C) = (A - B) \cup (A - C)$.**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
A - (B \cap C) &= \{x : (x \in A) \wedge (x \notin B \cap C)\} && \text{(definition of difference)} \\
&= \{x : (x \in A) \wedge \neg(x \in B \cap C)\} && \text{(notation)} \\
&= \{x : (x \in A) \wedge \neg(x \in B \wedge x \in C)\} && \text{(definition of } \cap\text{)} \\
&= \{x : (x \in A) \wedge (\neg(x \in B) \vee \neg(x \in C))\} && \text{(DeMorgan's law)} \\
&= \{x : (x \in A) \wedge (x \notin B \vee x \notin C)\} && \text{(notation)} \\
&= \{x : (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C)\} && \text{(distributive law)} \\
&= \{x : x \in A - B \vee x \in A - C\} && \text{(definition of difference)} \\
&= \{x : x \in (A - B) \cup (A - C)\} && \text{(definition of } \cup\text{)} \\
&= (A - B) \cup (A - C) && \text{(definition of set-builder)}
\end{aligned}$$

□

**14. If $A, B$ and $C$ are sets, then $(A \cup B) - C = (A - C) \cup (B - C)$.**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
(A \cup B) - C &= \{x : x \in (A \cup B) \wedge (x \notin C)\} && \text{(definition of difference)} \\
&= \{x : (x \in A \vee x \in B) \wedge (x \notin C)\} && \text{(definition of } \cup\text{)} \\
&= \{x : (x \in A \wedge x \notin C) \vee (x \in B \wedge x \notin C)\} && \text{(distributive law)} \\
&= \{x : x \in A - C \vee x \in B - C\} && \text{(definition of difference)} \\
&= \{x : x \in (A - C) \cup (B - C)\} && \text{(definition of } \cup\text{)} \\
&= (A - C) \cup (B - C) && \text{(definition of set-builder)}
\end{aligned}$$

□

**16. If $A, B$ and $C$ are sets, then $A \times (B \cup C) = (A \times B) \cup (A \times C)$.**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
A \times (B \cup C) &= \{(x, y) : x \in A \wedge y \in B \cup C\} && \text{(definition of } \times\text{)} \\
&= \{(x, y) : x \in A \wedge (y \in B \vee y \in C)\} && \text{(definition of } \cup\text{)} \\
&= \{(x, y) : (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)\} && \text{(distributive law)} \\
&= \{(x, y) : (x, y) \in A \times B \vee (x, y) \in A \times C\} && \text{(definition of } \times\text{)} \\
&= \{(x, y) : (x, y) \in (A \times B) \cup (A \times C)\} && \text{(definition of } \cup\text{)} \\
&= (A \times B) \cup (A \times C) && \text{(definition of set-builder)}
\end{aligned}$$

□

**18. If $A, B$ and $C$ are sets, then $A \times (B - C) = (A \times B) - (A \times C)$.**

*Proof.* Observe the following sequence of equalities.

$$
\begin{aligned}
A \times (B - C) &= \{(x, y) : x \in A \wedge y \in B - C\} && \text{(definition of } \times\text{)} \\
&= \{(x, y) : x \in A \wedge (y \in B \wedge y \notin C)\} && \text{(definition of difference)} \\
&= \{(x, y) : (x \in A \wedge y \in B \wedge x \notin A) && \text{(first disjunct is false)} \\
&\qquad \vee (x \in A \wedge y \in B \wedge y \notin C)\} && \\
&= \{(x, y) : (x \in A \wedge y \in B) \wedge (x \notin A \vee y \notin C)\} && \text{(distributive law)} \\
&= \{(x, y) : x \in A \wedge y \in B \wedge \neg(x \in A \wedge y \in C)\} && \text{(DeMorgan's law)} \\
&= \{(x, y) : (x, y) \in A \times B \wedge (x, y) \notin A \times C\} && \text{(definition of } \times\text{)} \\
&= \{(x, y) : (x, y) \in (A \times B) - (A \times C)\} && \text{(definition of difference)} \\
&= (A \times B) - (A \times C) && \text{(definition of set-builder)}
\end{aligned}
$$

$\square$

**20. Prove that $\{9^n : n \in \mathbb{Q}\} = \{3^n : n \in \mathbb{Q}\}$.**

*Proof.* First we prove $\{9^n : n \in \mathbb{Q}\} \subseteq \{3^n : n \in \mathbb{Q}\}$. Assume $a \in \{9^n : n \in \mathbb{Q}\}$. This means $a = 9^n$ for some $n \in \mathbb{Q}$. Thus $a = 9^n = (3^2)^n = 3^{2n}$. Since $2n \in \mathbb{Q}$, we have $a \in \{3^n : n \in \mathbb{Q}\}$. Therefore $\{9^n : n \in \mathbb{Q}\} \subseteq \{3^n : n \in \mathbb{Q}\}$.

Now we prove $\{3^n : n \in \mathbb{Q}\} \subseteq \{9^n : n \in \mathbb{Q}\}$. Assume $a \in \{3^n : n \in \mathbb{Q}\}$. This means $a = 3^n$ for some $n \in \mathbb{Q}$. Thus $a = 3^n = (9^{1/2})^n = 9^{n/2}$. Since $n/2 \in \mathbb{Q}$, we have $a \in \{9^n : n \in \mathbb{Q}\}$. Therefore $\{3^n : n \in \mathbb{Q}\} \subseteq \{9^n : n \in \mathbb{Q}\}$.

Therefore, since $\{9^n : n \in \mathbb{Q}\} \subseteq \{3^n : n \in \mathbb{Q}\}$ and $\{3^n : n \in \mathbb{Q}\} \subseteq \{9^n : n \in \mathbb{Q}\}$, it follows that $\{9^n : n \in \mathbb{Q}\} = \{3^n : n \in \mathbb{Q}\}$. $\square$

**22. Let $A$ and $B$ be sets. Prove that $A \subseteq B$ if and only if $A \cap B = A$.**

*Proof.* First we prove if $A \subseteq B$, then $A \cap B = A$.
Suppose $A \subseteq B$. Then every element of $A$ is also an element of $B$. Thus an element is in both $A$ and $B$ exactly when is in $A$. Therefore $A \cap B = A$.

Conversely, we prove if $A \cap B = A$, then $A \subseteq B$.
Suppose $A \cap B = A$ and let $x \in A$. Since $A = A \cap B$, it follows that $x \in B$. Thus every element of $A$ is also an element of $B$. Therefore $A \subseteq B$. $\square$